

Take home Quiz :-

1) $\frac{dy}{dt} = v(t) = 2e^{-t}$

$$\frac{dCA}{dt} = v_0 CA_{in} - KCAV$$

$$V \frac{dCA}{dt} + CA \frac{dV}{dt} = v_0 CA_{in} - KCAV$$

$$\frac{dV}{dt} = v_0$$

$$V \frac{dCA}{dt} + CA v_0 = v_0 CA_{in} - KCAV$$

$$\frac{dCA}{dt} = \frac{v_0}{V} (CA_{in} - CA) - KCA$$

$$\frac{dCB}{dt} = rBV = KCAV$$

$$V \frac{dCB}{dt} + CB v_0 = KCAV$$

$$\frac{dCB}{dt} = \frac{v_0 CB + KCA}{V}$$

2)

The screenshot shows a Python IDE with the following code in a file named 'untitled0.py':

```

import numpy as np
from scipy.integrate import odeint
import matplotlib.pyplot as plt

# Constants
CA_in = 0.001
V0 = 0.001
K = 0.001
rB = 0.001

# Initial conditions
CA = 0
CB = 0
V = 0

# Time span
t = np.linspace(0, 10, 100)

def model(y, t):
    CA, CB, V = y
    dCA = (V0*(CA_in - CA) - K*CA*V)
    dCB = (rB*V)
    dV = V0
    return [dCA, dCB, dV]

# Solve
sol = odeint(model, y0, t)

# Plot
plt.figure()
plt.plot(t, sol[:,0], label='CA')
plt.plot(t, sol[:,1], label='CB')
plt.plot(t, sol[:,2], label='V')
plt.legend()
plt.show()

```

The Variable Explorer window shows the following variables:

Name	Type	Size	Value
CA	Array of float64	[200]	[0.00231245 0.00427563 ... 0.0007923]
CA_in	float	1	0.001
CB	Array of float64	[200]	[0.00000000e+00 2.35178399e-05 8.79875100e-05 ... 0.00000000e+00]
CB	float	1	0.001
K	float	1	0.001
sol	Array of float64	[200, 3]	[[0.00000000e+00 0.00000000e+00 2.00000000e+00] ... [0.00231245e+00 2.35178399e-05 8.79875100e-05]]
t	Array of float64	[200]	[0.00000000e+00 0.00000000e+00 0.00000000e+00 ... 0.00000000e+00]
V	Array of float64	[200]	[0.00000000e+00 0.00000000e+00 0.00000000e+00 ... 0.00000000e+00]
V0	float	1	0.001

